Rethabile Mokoena

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About Me

Passionate and solution-driven computer science student at the University of Pretoria with expertise in **Java Spring Boot**, **AWS**, **PostgreSQL**, **Docker**, **Python**, and full-stack development. Experienced in building scalable backend systems with **RESTful APIs**, **Node.js**, and real-time features using **WebSockets**. Strong background in **machine learning** with Scikit-learn, Pandas, and Flask for AI-powered applications. Proven ability to develop efficient, secure software solutions from intelligent APIs to real-time web applications. Portfolio: rethamokoena.github.io/profile

Education

University of Pretoria — Pretoria, GP
Bachelor of Science in Computer Science — *Feb 2023 – Present*

Projects

Future Feed (Backend Engineer) | *Java Spring Boot, PostgreSQL, AWS, Docker, Oauth2.0* — Jan 2025 – 25 Oct

- *EPI-USE Labs (COS 301 Capstone Project)* — github.com/COS301-SE-2025/Future-Feed
- Architected a scalable backend supporting 1,000+ concurrent users with sub-500 ms feed generation.
- Developed RESTful APIs for user management, post interactions, and AI integrations (ChatGPT, Gemini).
- Implemented a rule-based feed ranking system (user-defined topic weighting) and JWT-based security.
- Reduced API costs by 40% through caching and optimized database queries.

Real-Time Auction System | *Node.js, Socket.io, PHP, MySQL, HTML, CSS, JavaScript* — Feb 2024 – Jun 2024

- github.com/RethaMokoena/Auction-System
- Built a full-stack auction platform enabling real-time bidding with WebSocket updates.
- Implemented secure authentication using password hashing and session management.
- Automated auction state transitions (Waiting → Active → Closed) via scheduled logic for seamless bidding.
- Developed an admin dashboard for real-time monitoring and user management.

Malicious URL Detection System | *Python, Scikit-learn, Pandas, Flask, Chrome Extension* — Ongoing

- github.com/RethaMokoena/Malicious-Detection-System
- Developed a machine learning pipeline to classify phishing and malicious URLs using Random Forest.
- Designed modular OOP framework with components for feature extraction, dataset loading, and model training.
- Engineered handcrafted features (special characters, URL length, redirection flags) to improve model accuracy.
- Integrated model persistence (joblib) and Chrome extension for real-time malicious link detection.

Technical Skills

Languages: Java, Python, C, C++, C#, SQL (PostgreSQL, MySQL), JavaScript, PHP, HTML5, CSS3, R, LaTeX
Frameworks & Platforms: Spring Boot, Node.js, Express.js, React, Flask, FastAPI, WordPress, JUnit, Material-UI, OpenGL, Socket.io, Bootstrap
Developer Tools: Git, Docker, Postman, Swagger, Maven, Gradle, VS Code, IntelliJ, PyCharm, Eclipse
Libraries & APIs: Pandas, NumPy, Matplotlib, JWT, WebSockets, OAuth 2.0, OpenAI API, Gemini API, OpenCV
Database Systems: PostgreSQL, MySQL, SQLite, MongoDB (basic)
Cloud & DevOps: AWS EC2/S3 (basic), GitHub Actions, WSL
Other: Object-Oriented Programming, Test-Driven Development, UML, Agile (Scrum), Data Structures & Algorithms